Dusk vs V1 Benchmark: Verified Replay, No Certified Winner

DecisionGrade technical report for the five-market selected panel at slot 440459364.

Technical summary

The historical V1 reconstruction is exact for the selected panel, but the Dusk comparison has no certifiable winner. All 139,812 canonical operations replayed into 139,825 transitions with zero raw-unit differences and zero conservation failures. Of 32 deterministic Dusk profiles, 31 fail exact curve geometry and one initializes all five markets; that profile still lacks complete scoring evidence because 77 LP withdrawals are unavailable after rejected deposits and 1,695 continuous hourly USD keys are absent. The correct verdict is **no DecisionGrade comparative claim**, not a forced estimate.

Canonical operations

139.81K

Canonical V1 operations replayed through the native model.

Source: Exact V1 replay evidence

Table: v1-reconciliation.json

Raw-unit differences

0

Nonzero integer-state differences across exact comparisons.

Source: Exact V1 replay evidence

Table: v1-reconciliation.json

Conservation failures

0

Failed accounting conservation identities.

Source: Exact V1 replay evidence

Table: v1-reconciliation.json

Certified scored profiles

0

Profiles with complete evidence for a comparative score.

Source: Dusk calibration preflight

Tables: fixed-flow arm evidence, intent arm evidence

The panel represents lending better than spot activity

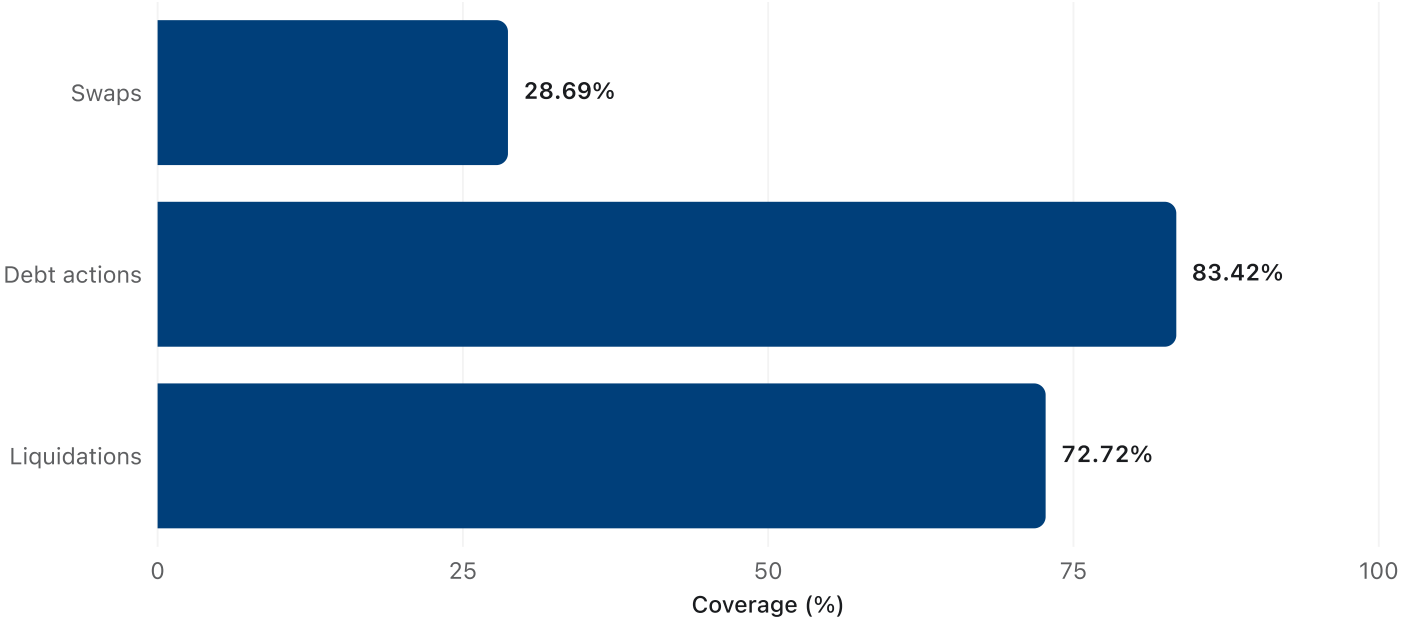
The five selected markets cover **83.42% of visible debt actions** and **72.72% of liquidations**, but only **28.69% of swaps**. This panel is decision-useful for lending and liquidation behavior; it is not a full-universe claim about spot execution.

Source: Selected-panel representativeness

Table: cohort-representativeness.json

Selected-panel activity coverage

Source: Selected-panel representativeness
Table: cohort-representativeness.json



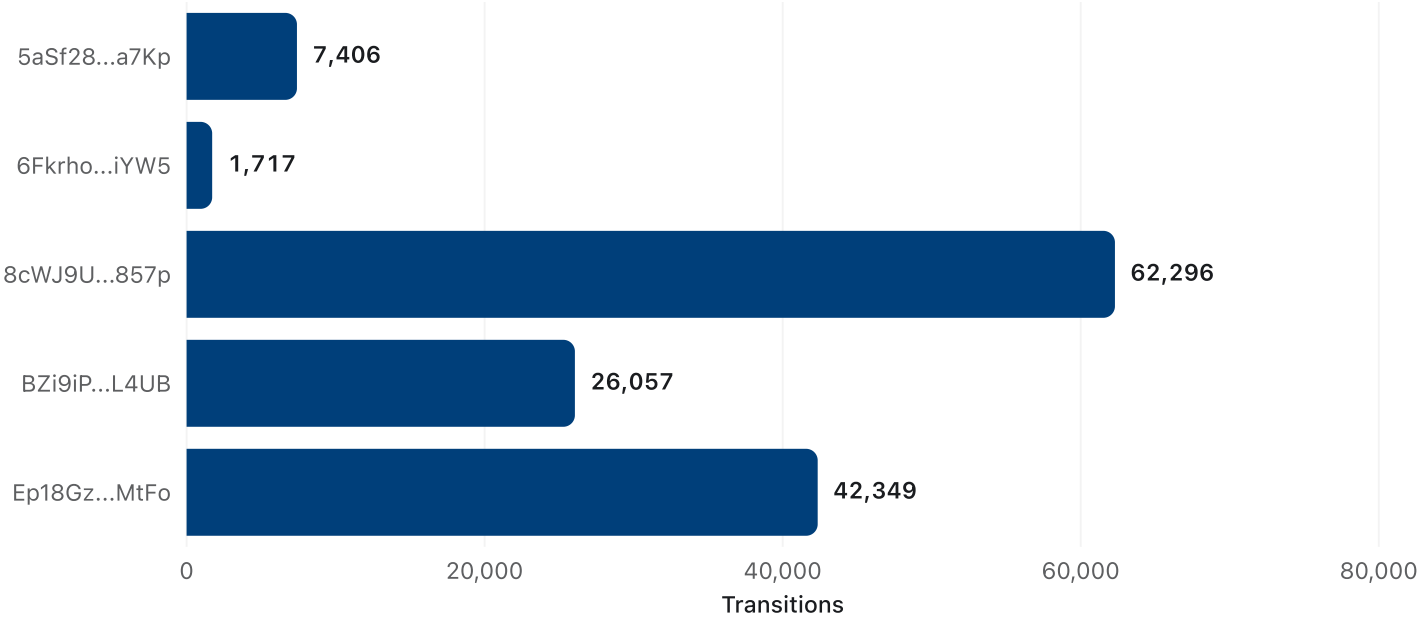
The V1 evidence path is internally exact

Across five markets, replay performed **5,599,341 exact raw-unit comparisons** and **838,950 conservation checks** without a failure. This validates the historical state reconstruction and accounting layer; it does not validate the missing terminal snapshot or deployed decoder-binary certification.

Source: Exact V1 replay evidence
Table: v1-reconciliation.json

Exact replay transitions by selected market

Source: Exact V1 replay evidence
Table: v1-reconciliation.json



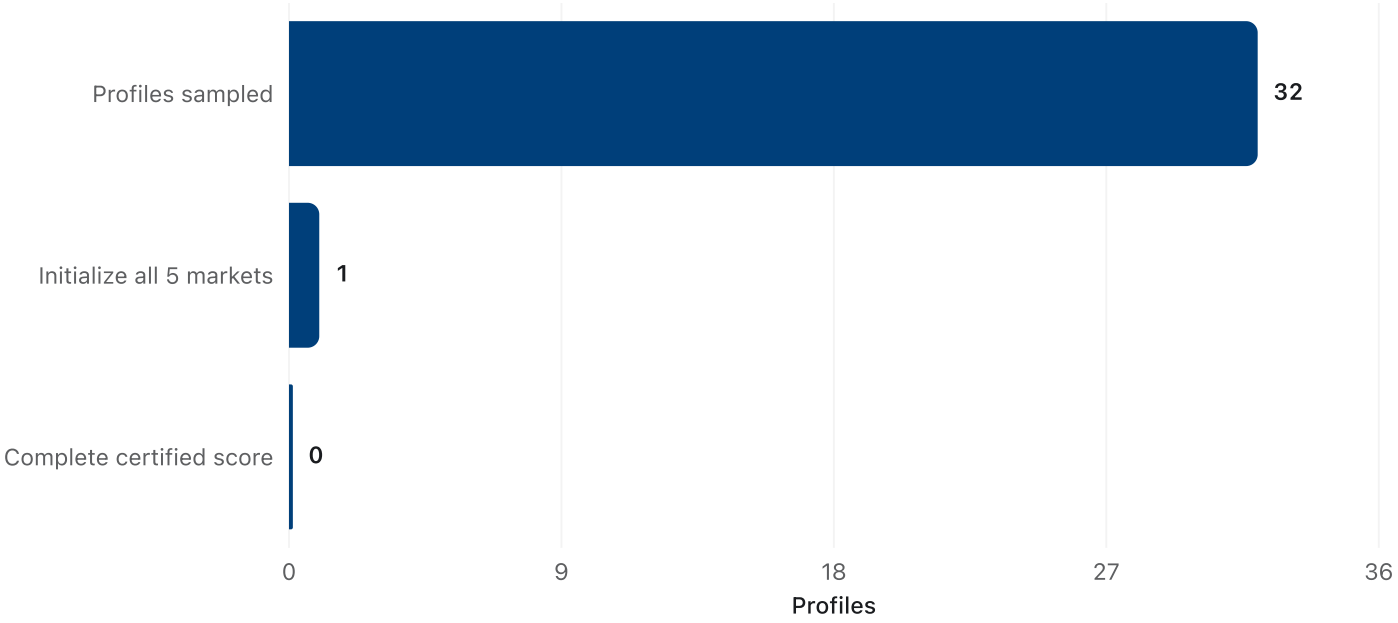
Calibration ends with zero decision-ready profiles

Only profile `d36b6a...e0d3b` initializes all five markets. It replays both fixed-flow and intent arms, but neither arm produces a complete candidate score. Missing inputs are retained as unavailable: no price interpolation, current-price substitution, or assumed LP ownership is used.

Source: Dusk calibration preflight
Tables: fixed-flow arm evidence, intent arm evidence

Calibration profile viability

Source: Dusk calibration preflight
Tables: fixed-flow arm evidence, intent arm evidence



Scope, data, and metric definitions

Cohort. Five explicitly selected, mint-bound V1 markets through slot 440,459,364. **Replay denominator.** Every canonical operation in the sealed selected-scope history. **Coverage denominator.** All activity rows in the visible-market catalog. **Winner requirement.** A protocol-wide shared Dusk profile with complete training and validation evidence for safety, execution quality, stakeholder revenue, and borrowing capacity. **Evidence tier.** DecisionGrade, with typed decoder-revision and terminal-state limitations.

Final DecisionGrade gate results

Source: DecisionGrade phase gates

Table: v1-final-phase-gate.json

Gate	Status	Observed / expected	Blocking
Authoritative Block Time	pass	264,017 / 264,017	0
Canonical Order	pass	264,017 / 264,017	0
Decoder Revision Certification	fail	0 / 1	1
Economic Replay	pass	139,812 / 139,812	0
Historical Prices	pass	279,608 / 279,608	0
Historical Transfer Fees	pass	11 / 11	0
Source Agreement	pass	263,977 / 263,977	0
State Changing Decode	pass	160,749 / 160,749	0
Terminal State	fail	0 / 5	5
Transaction Inventory	pass	137,329 / 137,329	0
Versioned Config	pass	838,848 / 838,848	0

Methodology preserves exactness and unavailability

The benchmark combines database event discovery with proof-bound archival RPC artifacts, exact instruction/CPI normalization, chronological V1 native replay, a sealed 60/20/20 split, deterministic Dusk profile exploration, and two replay arms. Historical prices are accepted only as provider-bound observations for the exact required hour. Unsupported geometry is a typed profile rejection; other errors remain fatal. Missing external LP ownership and terminal raw account state are never synthesized.

The actual row-level data is retained

The result is not based on summary-only inputs. The run retains canonical operations and events, decoded transactions/instructions, execution provenance, historical prices, database/RPC reconciliation, the exact V1 replay ledger, Parquet mirrors, and 115,850 proof-bound block artifacts. Across these sources there are zero normalization gaps, transaction decode failures, unresolved archival blocks, raw-unit replay differences, or conservation failures. The separate `DATA_INDEX.md` provides direct file links, grains, quality status, and five-row samples.

Source: Row-level benchmark data catalog
Table: data-catalog.json

Actual retained benchmark datasets

Source: Row-level benchmark data catalog
Table: data-catalog.json

Dataset	Rows	Formats	Size
Canonical operations	139.81K	JSONL + PARQUET	359.5 MiB + 123.3 MiB
Canonical events	264.02K	JSONL + PARQUET	260.9 MiB + 89.1 MiB
V1 replay ledger	139.83K	JSONL + PARQUET	675.5 MiB + 90.7 MiB
Historical prices	22.07K	JSONL + PARQUET	3.9 MiB + 597.5 KiB
Decoded transactions	137.28K	JSONL	1.1 GiB
Decoded instructions	462.31K	JSONL	468.9 MiB
Execution provenance	137.28K	JSONL	1.4 GiB
Database events	895.28K	JSONL	543.1 MiB
DB/RPC reconciliation	263.98K	JSONL	74.4 MiB
Archived proof blocks	115.85K	JSON.ZST	11.0 GiB

Limitations prevent a comparative verdict

- Historical decoder dispatch is structurally bound to a candidate binary but not certified against every deployed revision.
- Terminal on-chain account snapshots were not captured, so certification remains limited.
- Seventy-seven LP withdrawals are not replayable after Dusk rejects their antecedent deposits.
- Continuous price coverage needed for capital-time scoring is incomplete by 1,695 mint-hour keys.
- The selected panel covers 28.69% of visible swaps and must not be described as the full spot universe.

Recommended next step: stop the expensive campaign

Publish this as a **no-winner benchmark result**. Do not spend more storage replaying invalid profiles or impute the missing state. A future rerun is justified only if Dusk defines how rejected V1 LP deposits map to later withdrawals and the data contract supplies continuous historical price marks plus a terminal snapshot.

Further questions

1. Should the product define a deterministic no-counterpart policy for withdrawals whose factual deposits Dusk rejected?
2. Is capital-time weighting required for the launch decision, or can an operation-weighted secondary metric be explicitly labeled?
3. Can deployed V1 program binaries and cutoff account snapshots be independently certified without reopening full-universe extraction?